

PROJECT REPORT

ADVENTURE WORKS

SQL DATA ANALYSIS PROJECT

PROJECT DOMAIN

SQL / Data Engineering

Project Name	:	**Adventure Works**
Assigned By	:	**Neha**
Domain	:	**SQL**
Completed By	:	**Aditya Kadu**

SQL Server | Data Analysis | Reporting

DAY1 TASK

1) .List every table in the SalesLT schema with its row count, using ONE query (hint: sys.tables + sys.partitions).

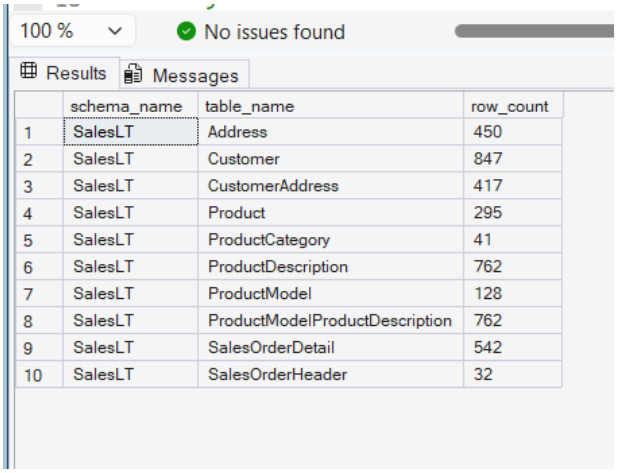
First, I select the information about tables from the SQL Server system view sys.tables.

Then, I join it with sys.partitions to get the row-count information.

I match both sources using object_id. After that,

I filter the results to include only the SalesLT schema and select the appropriate index types to avoid counting rows more than once.

Finally, I group the data by schema and table name, calculate the total row count, and sort the table names alphabetically.



	schema_name	table_name	row_count
1	SalesLT	Address	450
2	SalesLT	Customer	847
3	SalesLT	CustomerAddress	417
4	SalesLT	Product	295
5	SalesLT	ProductCategory	41
6	SalesLT	ProductDescription	762
7	SalesLT	ProductModel	128
8	SalesLT	ProductModelProductDescription	762
9	SalesLT	SalesOrderDetail	542
10	SalesLT	SalesOrderHeader	32

2. Find the primary key and foreign keys of every table (INFORMATION_SCHEMA and sys.foreign_keys).

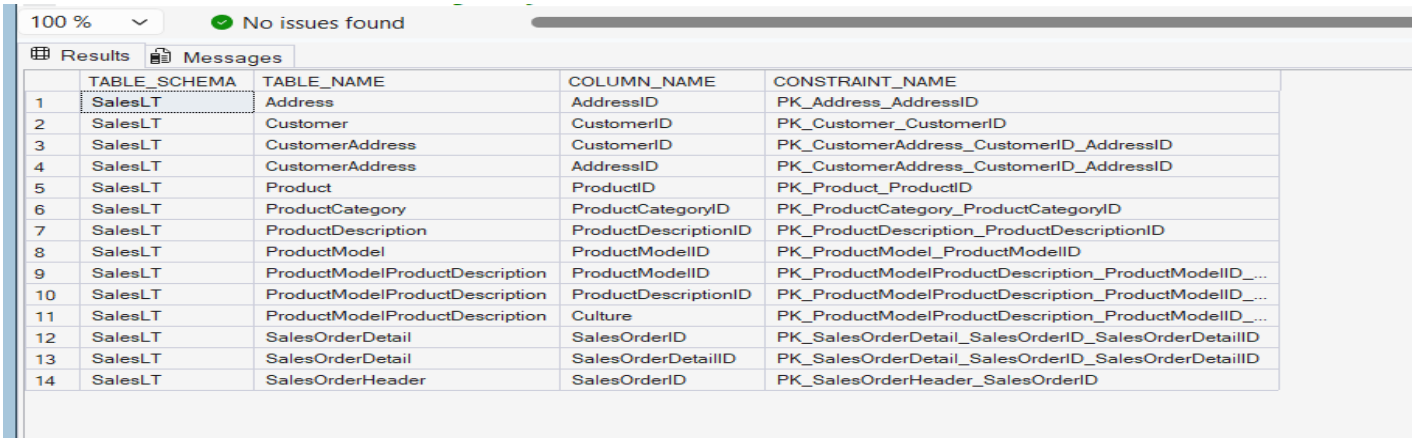
First, I select the AdventureWorksLT2022 database. Then, I retrieve schema names, table names,

column names, and constraint names from INFORMATION_SCHEMA.KEY_COLUMN_USAGE.

I use a subquery on INFORMATION_SCHEMA.TABLE_CONSTRAINTS to identify constraints of type Primary Key.

I filter the results to the SalesLT schema, and finally, I sort them by table name and column position.

This helps me identify the Primary Key columns in each table.



100 % No issues found

Results Messages

	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
1	SalesLT	Address	AddressID	PK_Address_AddressID
2	SalesLT	Customer	CustomerID	PK_Customer_CustomerID
3	SalesLT	CustomerAddress	CustomerID	PK_CustomerAddress_CustomerID_AddressID
4	SalesLT	CustomerAddress	AddressID	PK_CustomerAddress_CustomerID_AddressID
5	SalesLT	Product	ProductID	PK_Product_ProductID
6	SalesLT	ProductCategory	ProductCategoryID	PK_ProductCategory_ProductCategoryID
7	SalesLT	ProductDescription	ProductDescriptionID	PK_ProductDescription_ProductDescriptionID
8	SalesLT	ProductModel	ProductModelID	PK_ProductModel_ProductModelID
9	SalesLT	ProductModelProductDescription	ProductModelID	PK_ProductModelProductDescription_ProductModelID_...
10	SalesLT	ProductModelProductDescription	ProductDescriptionID	PK_ProductModelProductDescription_ProductModelID_...
11	SalesLT	ProductModelProductDescription	Culture	PK_ProductModelProductDescription_ProductModelID_...
12	SalesLT	SalesOrderDetail	SalesOrderID	PK_SalesOrderDetail_SalesOrderID_SalesOrderDetailID
13	SalesLT	SalesOrderDetail	SalesOrderDetailID	PK_SalesOrderDetail_SalesOrderID_SalesOrderDetailID
14	SalesLT	SalesOrderHeader	SalesOrderID	PK_SalesOrderHeader_SalesOrderID

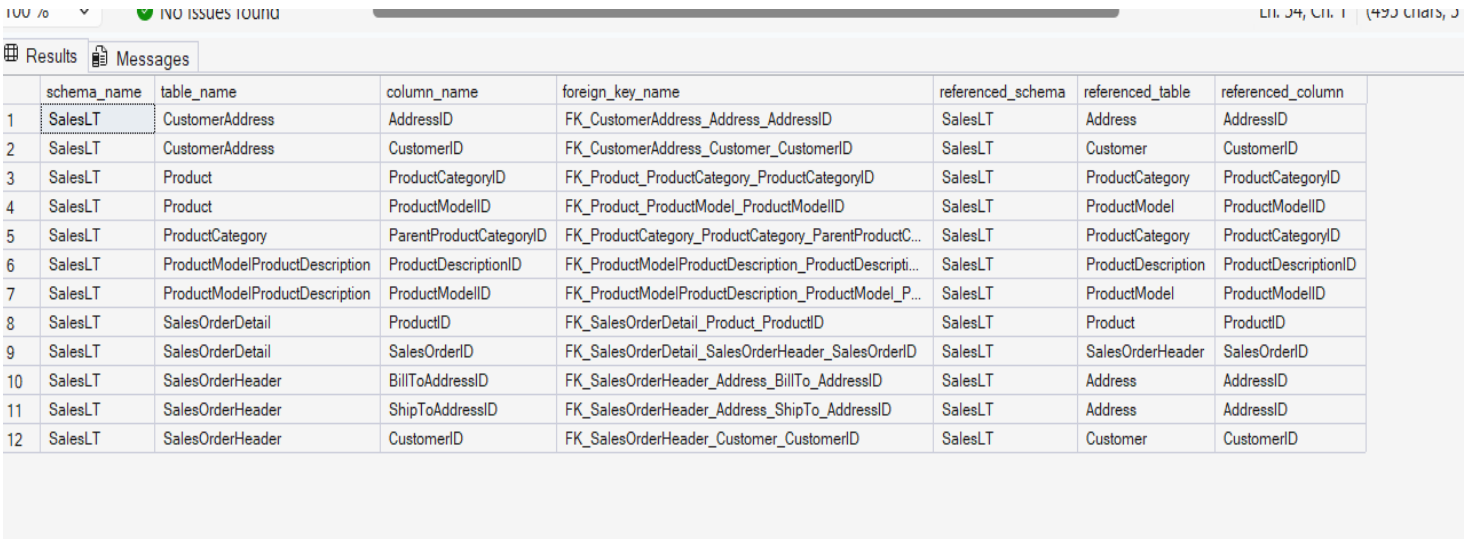
B. Find all Foreign Keys (FKs)

First, I access the AdventureWorksLT2022 database. Then, I retrieve Foreign Key information from the SQL Server system views.

I join the Foreign Key details with the table and column information to identify both sides of each relationship.

After that, I filter the results to include only the SalesLT schema. Finally, I display the Foreign Key name, the column containing it,

and the referenced table and column, and sort the results by table name.

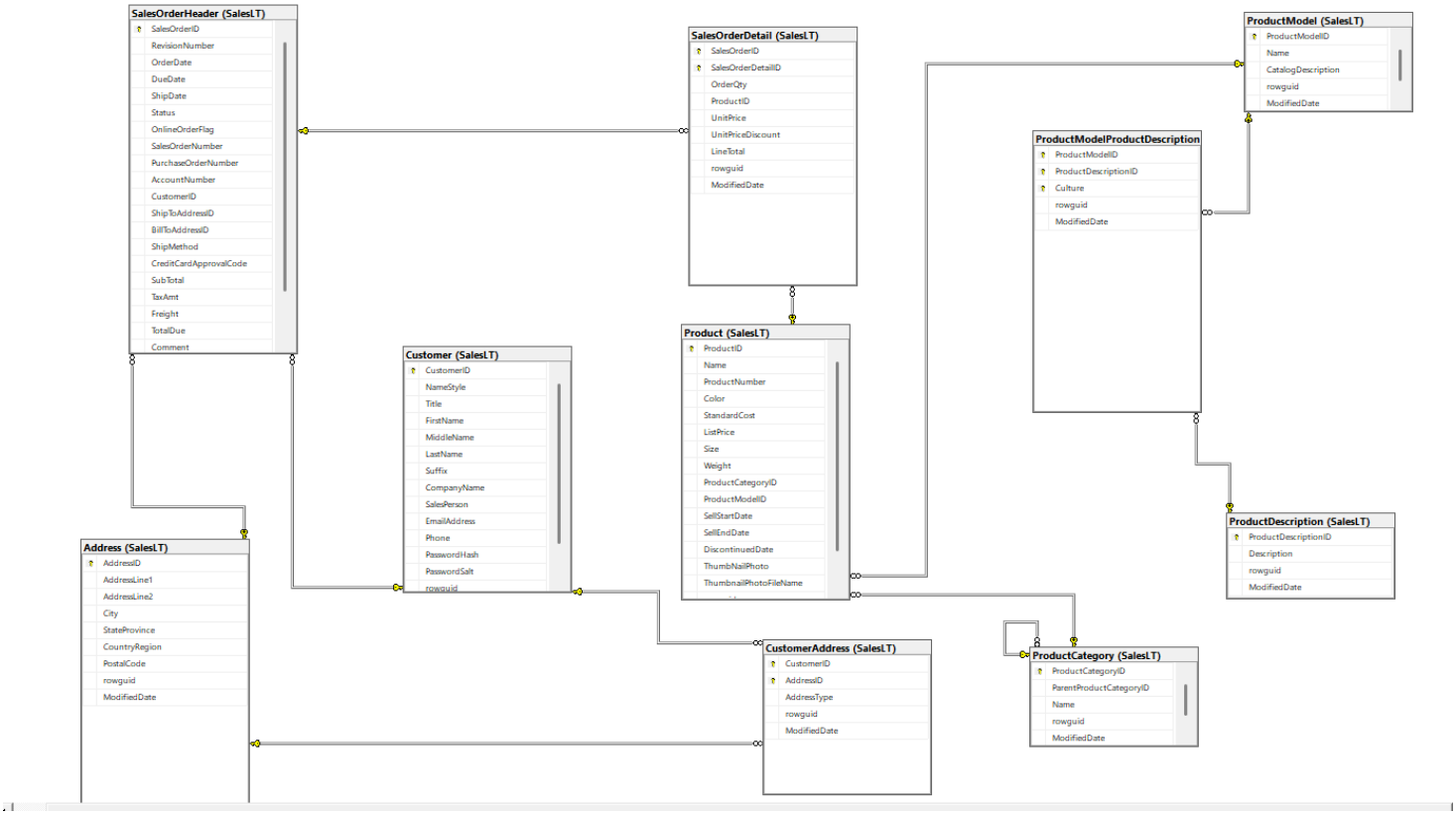


No issues found

Results Messages

	schema_name	table_name	column_name	foreign_key_name	referenced_schema	referenced_table	referenced_column
1	SalesLT	CustomerAddress	AddressID	FK_CustomerAddress_Address_AddressID	SalesLT	Address	AddressID
2	SalesLT	CustomerAddress	CustomerID	FK_CustomerAddress_Customer_CustomerID	SalesLT	Customer	CustomerID
3	SalesLT	Product	ProductCategoryID	FK_Product_ProductCategory_ProductCategoryID	SalesLT	ProductCategory	ProductCategoryID
4	SalesLT	Product	ProductModelID	FK_Product_ProductModel_ProductModelID	SalesLT	ProductModel	ProductModelID
5	SalesLT	ProductCategory	ParentProductCategoryID	FK_ProductCategory_ProductCategory_ParentProductC...	SalesLT	ProductCategory	ProductCategoryID
6	SalesLT	ProductModelProductDescription	ProductDescriptionID	FK_ProductModelProductDescription_ProductDescripti...	SalesLT	ProductDescription	ProductDescriptionID
7	SalesLT	ProductModelProductDescription	ProductModelID	FK_ProductModelProductDescription_ProductModel_P...	SalesLT	ProductModel	ProductModelID
8	SalesLT	SalesOrderDetail	ProductID	FK_SalesOrderDetail_Product_ProductID	SalesLT	Product	ProductID
9	SalesLT	SalesOrderDetail	SalesOrderID	FK_SalesOrderDetail_SalesOrderHeader_SalesOrderID	SalesLT	SalesOrderHeader	SalesOrderID
10	SalesLT	SalesOrderHeader	BillToAddressID	FK_SalesOrderHeader_Address_BillTo_AddressID	SalesLT	Address	AddressID
11	SalesLT	SalesOrderHeader	ShipToAddressID	FK_SalesOrderHeader_Address_ShipTo_AddressID	SalesLT	Address	AddressID
12	SalesLT	SalesOrderHeader	CustomerID	FK_SalesOrderHeader_Customer_CustomerID	SalesLT	Customer	CustomerID

3)Draw an ER diagram of the source tables (dbdiagram.io or draw.io).



Task 4A — Check NULL counts and distinct counts

Step 1: Check Customer data

This query performs a data-quality check on the Customer table. First, it counts the total customer records.

Then, it counts the non-NULL and NULL values in the FirstName, LastName, and EmailAddress columns.

Finally, it counts the distinct email addresses.

This helps us understand whether customer information is missing and how many unique email addresses are available.

Results | Messages

	total_customers	non_null_first_names	null_first_names	non_null_last_names	null_last_names	non_null_emails	null_emails	distinct_emails
1	847	847	0	847	0	847	0	440

Step 2: Check Product data

In Step 2, I performed a data-quality check on the Product table.

I counted the total number of products and checked for missing values in the Name,

ProductNumber, and ListPrice columns. I also counted the distinct product numbers to check how many unique product identifiers are available.

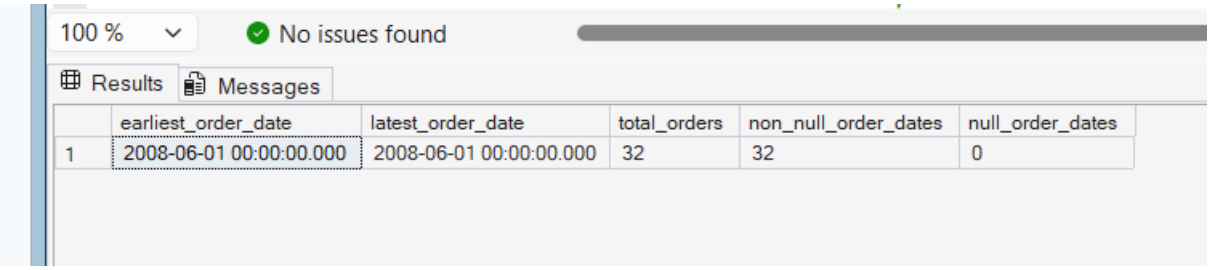
This helps us identify missing product information and potential duplicate product numbers.

100 % | No issues found | Ln: 1

Results | Messages

	total_products	non_null_names	null_names	non_null_product_numbers	null_product_numbers	non_null_list_prices	null_list_prices	distinct_product_numbers
1	295	295	0	295	0	295	0	295

Task 4B — Check minimum and maximum order dates



100 % No issues found

Results Messages

	earliest_order_date	latest_order_date	total_orders	non_null_order_dates	null_order_dates
1	2008-06-01 00:00:00.000	2008-06-01 00:00:00.000	32	32	0

Task 4C — Check zero or negative prices and quantities

I performed a data-quality check on the SalesOrderDetail table. First,

I counted the total order-detail records. Then, I checked UnitPrice and OrderQty for missing, zero and negative values using CASE WHEN conditions and SUM.

This helps identify potentially invalid pricing and quantity records before using the data for analysis.

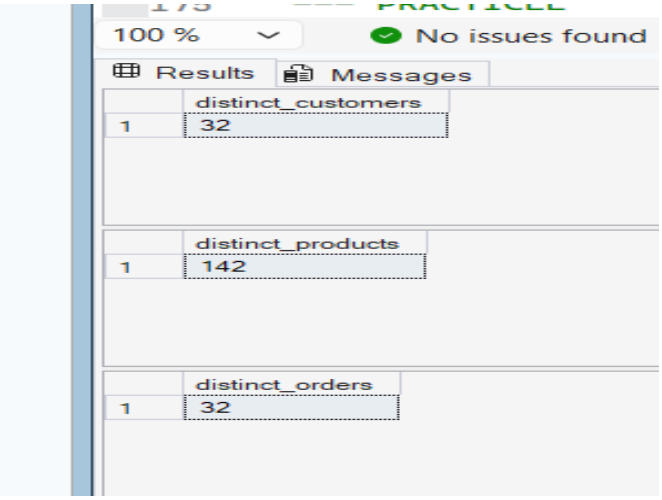


100 % No issues found

Results Messages

	total_order_details	null_unit_prices	zero_unit_prices	negative_unit_prices	null_quantities	zero_quantities	negative_quantities
1	542	0	0	0	0	0	0

Task 4D — Check distinct counts for important columns



100 % No issues found

Results Messages

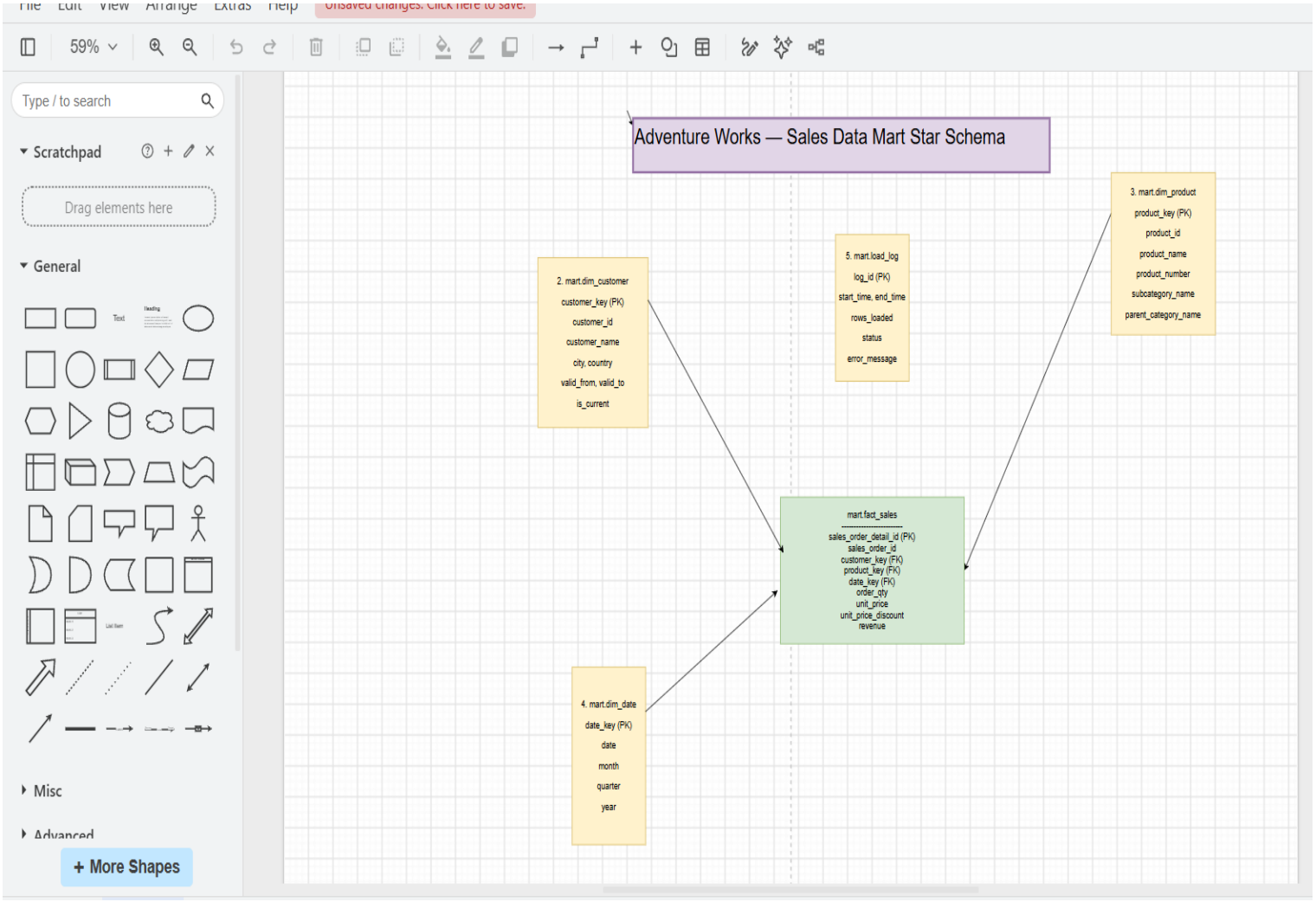
	distinct_customers
1	32

	distinct_products
1	142

	distinct_orders
1	32

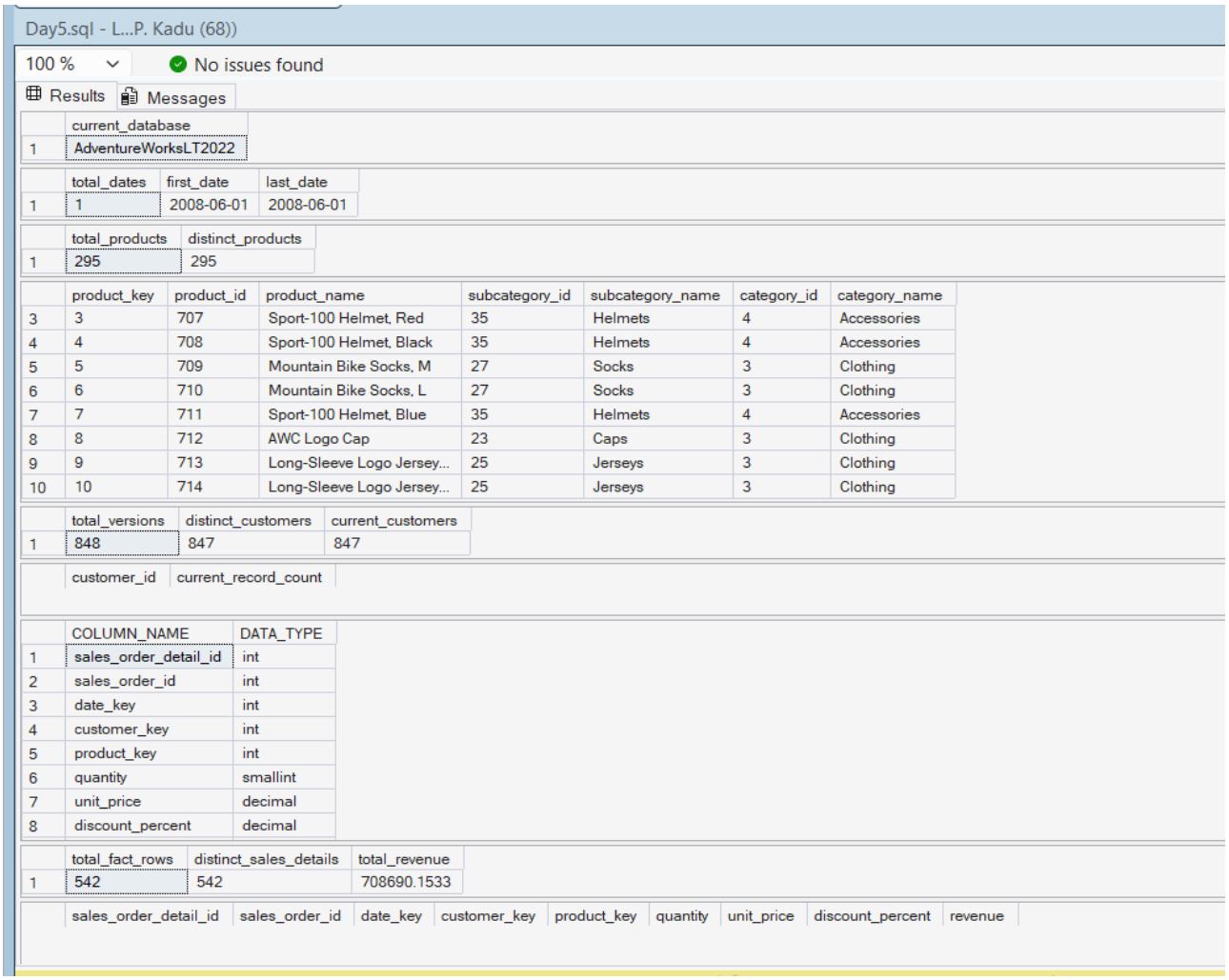
DAY 2 TASK

Draw the star schema.



DAY3TASK

1. CREATE SCHEMA mart; then write the DDL for all tables in 02_ddl.sql.
2. Load dim_date and dim_product using MERGE.
3. Load dim_customer as SCD Type 2: when city or country changes, close the old row (set valid_to, is_current = 0) and insert a new current row. Give the first version of each customer a valid_from of 1900-01-01, so old orders still find a match.
4. Load fact_sales so that it never duplicates: MERGE on the order-line ID, or delete and re-insert.
5. Put everything in mart.usp_load_sales_mart with TRY/CATCH and logging (pattern below).
6. Test SCD2: change one customer's city in SalesLT.Address, run the procedure again, and confirm that customer now has 2 rows with only one current.



Day5.sql - L...P. Kadu (68))

100 % No issues found

Results Messages

	current_database
1	AdventureWorksLT2022

	total_dates	first_date	last_date
1	1	2008-06-01	2008-06-01

	total_products	distinct_products
1	295	295

	product_key	product_id	product_name	subcategory_id	subcategory_name	category_id	category_name
3	3	707	Sport-100 Helmet, Red	35	Helmets	4	Accessories
4	4	708	Sport-100 Helmet, Black	35	Helmets	4	Accessories
5	5	709	Mountain Bike Socks, M	27	Socks	3	Clothing
6	6	710	Mountain Bike Socks, L	27	Socks	3	Clothing
7	7	711	Sport-100 Helmet, Blue	35	Helmets	4	Accessories
8	8	712	AWC Logo Cap	23	Caps	3	Clothing
9	9	713	Long-Sleeve Logo Jersey...	25	Jerseys	3	Clothing
10	10	714	Long-Sleeve Logo Jersey...	25	Jerseys	3	Clothing

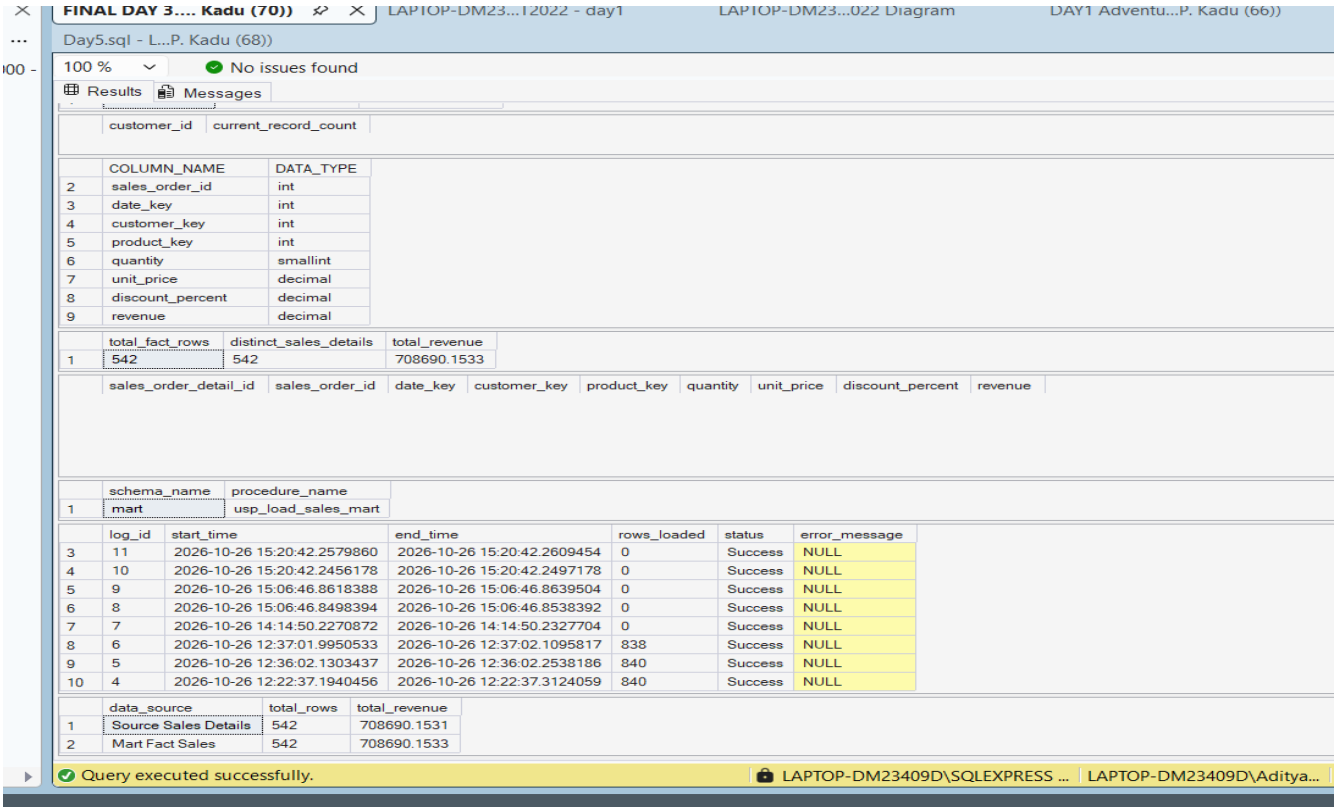
	total_versions	distinct_customers	current_customers
1	848	847	847

	customer_id	current_record_count

	COLUMN_NAME	DATA_TYPE
1	sales_order_detail_id	int
2	sales_order_id	int
3	date_key	int
4	customer_key	int
5	product_key	int
6	quantity	smallint
7	unit_price	decimal
8	discount_percent	decimal

	total_fact_rows	distinct_sales_details	total_revenue
1	542	542	708690.1533

	sales_order_detail_id	sales_order_id	date_key	customer_key	product_key	quantity	unit_price	discount_percent	revenue



FINAL DAY 3.... Kadu (70)) LAPTOP-DM23...T2022 - day1 LAPTOP-DM23...022 Diagram DAY1 Adventu...P. Kadu (66))

Day5.sql - L...P. Kadu (68))

100 % No issues found

Results Messages

	customer_id	current_record_count

	COLUMN_NAME	DATA_TYPE
2	sales_order_id	int
3	date_key	int
4	customer_key	int
5	product_key	int
6	quantity	smallint
7	unit_price	decimal
8	discount_percent	decimal
9	revenue	decimal

	total_fact_rows	distinct_sales_details	total_revenue
1	542	542	708690.1533

	sales_order_detail_id	sales_order_id	date_key	customer_key	product_key	quantity	unit_price	discount_percent	revenue

	schema_name	procedure_name
1	mart	usp_load_sales_mart

	log_id	start_time	end_time	rows_loaded	status	error_message
3	11	2026-10-26 15:20:42.2579860	2026-10-26 15:20:42.2609454	0	Success	NULL
4	10	2026-10-26 15:20:42.2456178	2026-10-26 15:20:42.2497178	0	Success	NULL
5	9	2026-10-26 15:06:46.8618388	2026-10-26 15:06:46.8639504	0	Success	NULL
6	8	2026-10-26 15:06:46.8498394	2026-10-26 15:06:46.8538392	0	Success	NULL
7	7	2026-10-26 14:14:50.2270872	2026-10-26 14:14:50.2327704	0	Success	NULL
8	6	2026-10-26 12:37:01.9950533	2026-10-26 12:37:02.1095817	838	Success	NULL
9	5	2026-10-26 12:36:02.1303437	2026-10-26 12:36:02.2538186	840	Success	NULL
10	4	2026-10-26 12:22:37.1940456	2026-10-26 12:22:37.3124059	840	Success	NULL

	data_source	total_rows	total_revenue
1	Source Sales Details	542	708690.1531
2	Mart Fact Sales	542	708690.1533

Query executed successfully. LAPTOP-DM23409D\SQLEXPRESS ... LAPTOP-DM23409D\Aditya...

DAY 4TASK

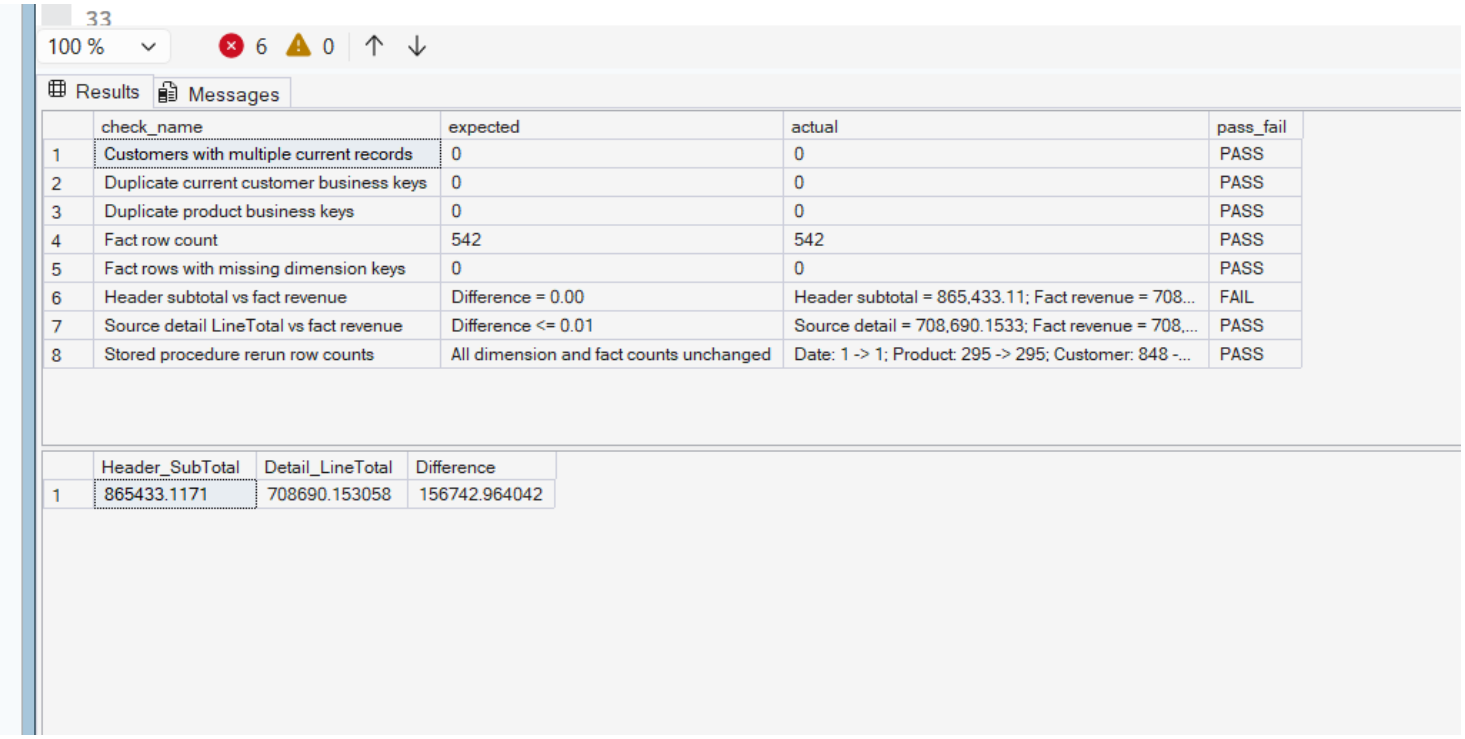
Output by end of day

- **04_dq_checks.sql with all checks passing, and a screenshot of the results table**

- **1. Why did this test fail?**
- Your output shows:

Column	Actual value
Header SubTotal	865,433.1171
Detail LineTotal	708,690.153058
Difference	156,742.964042

- The test expects the difference between the header subtotal and fact revenue to be `0.00`, but the actual difference is `156,742.96`.
-



33

100 % 6 0

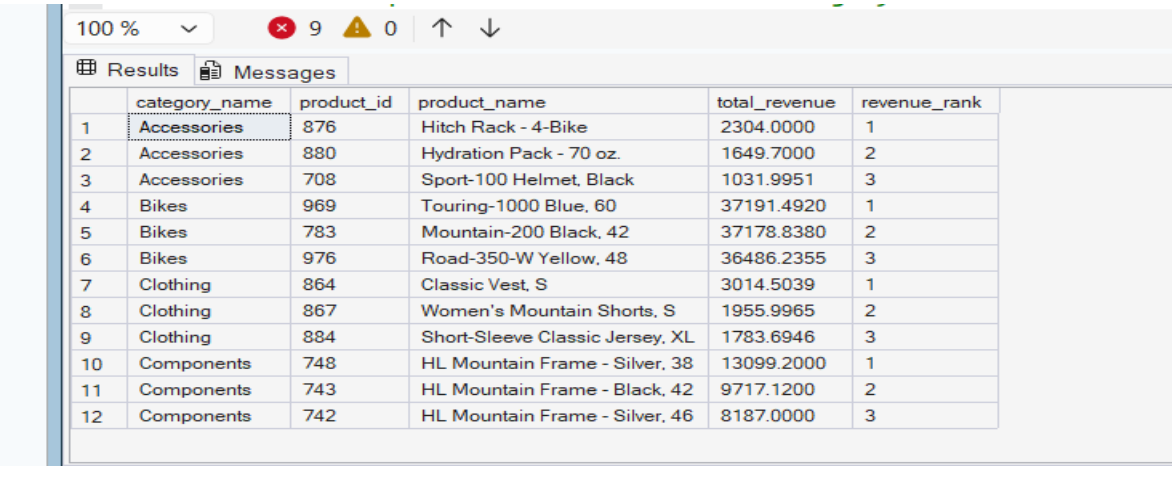
Results Messages

	check_name	expected	actual	pass_fail
1	Customers with multiple current records	0	0	PASS
2	Duplicate current customer business keys	0	0	PASS
3	Duplicate product business keys	0	0	PASS
4	Fact row count	542	542	PASS
5	Fact rows with missing dimension keys	0	0	PASS
6	Header subtotal vs fact revenue	Difference = 0.00	Header subtotal = 865,433.11; Fact revenue = 708...	FAIL
7	Source detail LineTotal vs fact revenue	Difference <= 0.01	Source detail = 708,690.1533; Fact revenue = 708,...	PASS
8	Stored procedure rerun row counts	All dimension and fact counts unchanged	Date: 1 -> 1; Product: 295 -> 295; Customer: 848 -...	PASS

	Header_SubTotal	Detail_LineTotal	Difference
1	865433.1171	708690.153058	156742.964042

DAY5TASK

1)Top 3 products by revenue within each category (DENSE_RANK). Explain why ROW_NUMBER could give a different answer.

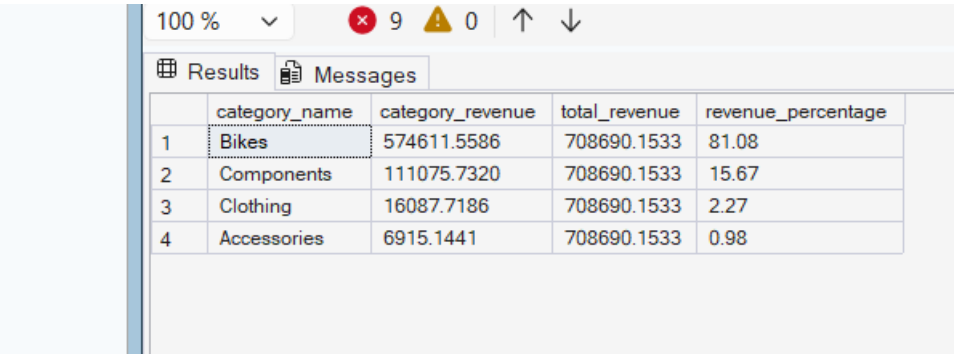


100 % 9 0

Results Messages

	category_name	product_id	product_name	total_revenue	revenue_rank
1	Accessories	876	Hitch Rack - 4-Bike	2304.0000	1
2	Accessories	880	Hydration Pack - 70 oz.	1649.7000	2
3	Accessories	708	Sport-100 Helmet, Black	1031.9951	3
4	Bikes	969	Touring-1000 Blue, 60	37191.4920	1
5	Bikes	783	Mountain-200 Black, 42	37178.8380	2
6	Bikes	976	Road-350-W Yellow, 48	36486.2355	3
7	Clothing	864	Classic Vest, S	3014.5039	1
8	Clothing	867	Women's Mountain Shorts, S	1955.9965	2
9	Clothing	884	Short-Sleeve Classic Jersey, XL	1783.6946	3
10	Components	748	HL Mountain Frame - Silver, 38	13099.2000	1
11	Components	743	HL Mountain Frame - Black, 42	9717.1200	2
12	Components	742	HL Mountain Frame - Silver, 46	8187.0000	3

2) Each category's share of total revenue (SUM() OVER ()).

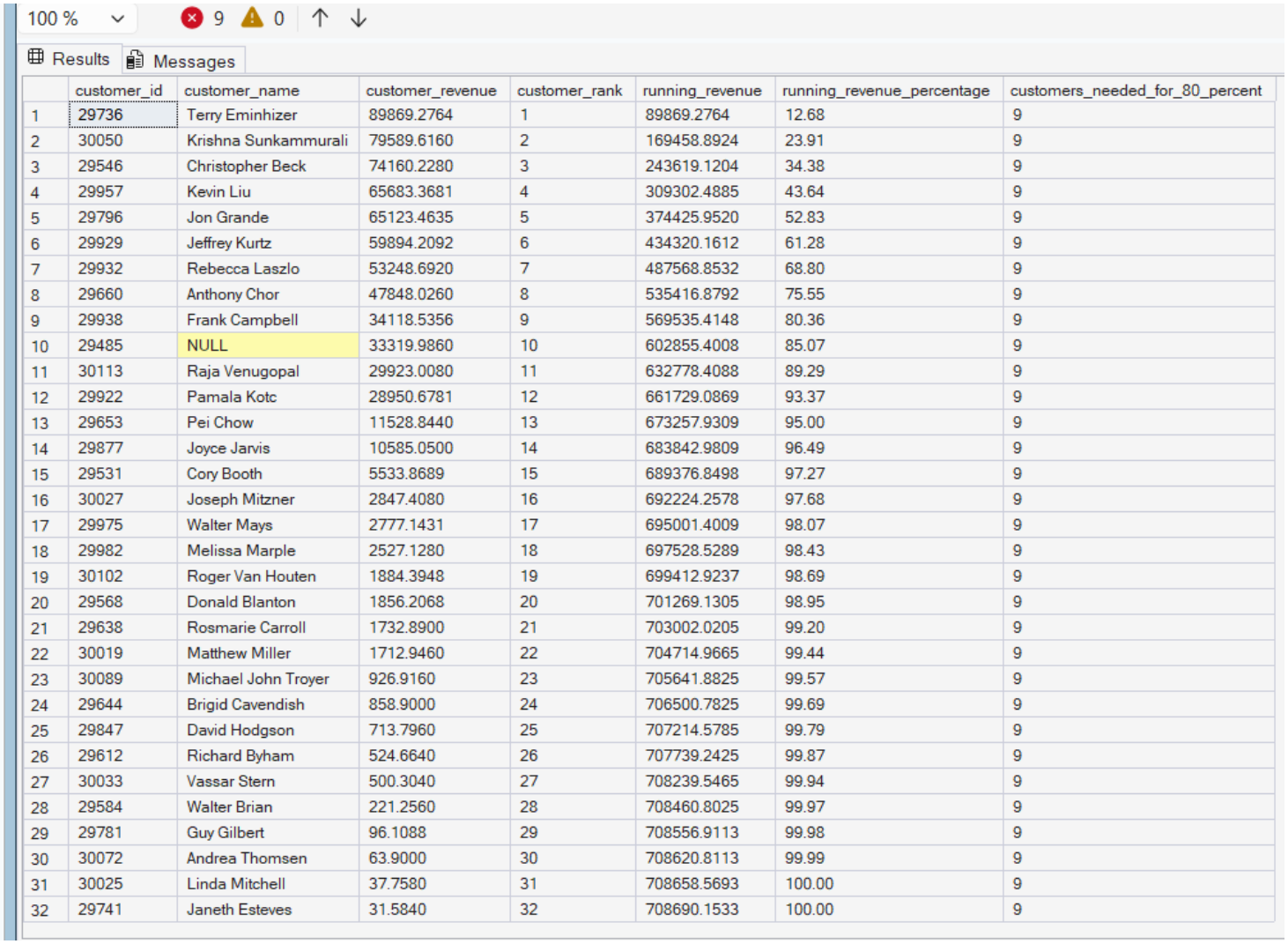


100 % 9 0

Results Messages

	category_name	category_revenue	total_revenue	revenue_percentage
1	Bikes	574611.5586	708690.1533	81.08
2	Components	111075.7320	708690.1533	15.67
3	Clothing	16087.7186	708690.1533	2.27
4	Accessories	6915.1441	708690.1533	0.98

3)Customer Pareto: running % of revenue by customer, highest first. How many customers make up 80% of revenue?



100 % 9 0

Results Messages

	customer_id	customer_name	customer_revenue	customer_rank	running_revenue	running_revenue_percentage	customers_needed_for_80_percent
1	29736	Terry Eminhizer	89869.2764	1	89869.2764	12.68	9
2	30050	Krishna Sunkammurali	79589.6160	2	169458.8924	23.91	9
3	29546	Christopher Beck	74160.2280	3	243619.1204	34.38	9
4	29957	Kevin Liu	65683.3681	4	309302.4885	43.64	9
5	29796	Jon Grande	65123.4635	5	374425.9520	52.83	9
6	29929	Jeffrey Kurtz	59894.2092	6	434320.1612	61.28	9
7	29932	Rebecca Laszlo	53248.6920	7	487568.8532	68.80	9
8	29660	Anthony Chor	47848.0260	8	535416.8792	75.55	9
9	29938	Frank Campbell	34118.5356	9	569535.4148	80.36	9
10	29485	NULL	33319.9860	10	602855.4008	85.07	9
11	30113	Raja Venugopal	29923.0080	11	632778.4088	89.29	9
12	29922	Pamala Kotc	28950.6781	12	661729.0869	93.37	9
13	29653	Pei Chow	11528.8440	13	673257.9309	95.00	9
14	29877	Joyce Jarvis	10585.0500	14	683842.9809	96.49	9
15	29531	Cory Booth	5533.8689	15	689376.8498	97.27	9
16	30027	Joseph Mitzner	2847.4080	16	692224.2578	97.68	9
17	29975	Walter Mays	2777.1431	17	695001.4009	98.07	9
18	29982	Melissa Marple	2527.1280	18	697528.5289	98.43	9
19	30102	Roger Van Houten	1884.3948	19	699412.9237	98.69	9
20	29568	Donald Blanton	1856.2068	20	701269.1305	98.95	9
21	29638	Rosmarie Carroll	1732.8900	21	703002.0205	99.20	9
22	30019	Matthew Miller	1712.9460	22	704714.9665	99.44	9
23	30089	Michael John Troyer	926.9160	23	705641.8825	99.57	9
24	29644	Brigid Cavendish	858.9000	24	706500.7825	99.69	9
25	29847	David Hodgson	713.7960	25	707214.5785	99.79	9
26	29612	Richard Byham	524.6640	26	707739.2425	99.87	9
27	30033	Vassar Stern	500.3040	27	708239.5465	99.94	9
28	29584	Walter Brian	221.2560	28	708460.8025	99.97	9
29	29781	Guy Gilbert	96.1088	29	708556.9113	99.98	9
30	30072	Andrea Thomsen	63.9000	30	708620.8113	99.99	9
31	30025	Linda Mitchell	37.7580	31	708658.5693	100.00	9
32	29741	Janeth Esteves	31.5840	32	708690.1533	100.00	9

4)Each customer's revenue rank within their country.

Results Messages

	customer_id	customer_name	country	customer_revenue	country_revenue_rank
1	29485	NULL	NULL	33319.9860	1
2	29736	Terry Eminhizer	United Kingdom	89869.2764	1
3	30050	Krishna Sunkammurali	United Kingdom	79589.6160	2
4	29546	Christopher Beck	United Kingdom	74160.2280	3
5	29796	Jon Grande	United Kingdom	65123.4635	4
6	29932	Rebecca Laszlo	United Kingdom	53248.6920	5
7	30113	Raja Venugopal	United Kingdom	29923.0080	6
8	29922	Pamala Kotc	United Kingdom	28950.6781	7
9	30102	Roger Van Houten	United Kingdom	1884.3948	8
10	30019	Matthew Miller	United Kingdom	1712.9460	9
11	29644	Brigid Cavendish	United Kingdom	858.9000	10
12	29612	Richard Byham	United Kingdom	524.6640	11
13	30072	Andrea Thomsen	United Kingdom	63.9000	12
14	30025	Linda Mitchell	United Kingdom	37.7580	13
15	29741	Janeth Esteves	United Kingdom	31.5840	14
16	29957	Kevin Liu	United States	65683.3681	1
17	29929	Jeffrey Kurtz	United States	59894.2092	2
18	29660	Anthony Chor	United States	47848.0260	3
19	29938	Frank Campbell	United States	34118.5356	4
20	29653	Pei Chow	United States	11528.8440	5
21	29877	Joyce Jarvis	United States	10585.0500	6
22	29531	Cory Booth	United States	5533.8689	7
23	30027	Joseph Mitzner	United States	2847.4080	8
24	29975	Walter Mays	United States	2777.1431	9
25	29982	Melissa Marple	United States	2527.1280	10
26	29568	Donald Blanton	United States	1856.2068	11
27	29638	Rosmarie Carroll	United States	1732.8900	12
28	30089	Michael John Troyer	United States	926.9160	13
29	29847	David Hodgson	United States	713.7960	14
30	30033	Vassar Stern	United States	500.3040	15
31	29584	Walter Brian	United States	221.2560	16
32	29781	Guy Gilbert	United States	96.1088	17

DE-101 – Q5: Month-over-month revenue and 3-month moving average

Script: 05_business_questions.sql (run block by block in SSMS)

1. Source check

AdventureWorksLT2022.SalesLT.SalesOrderHeader (query result):

total_orders	first_order	last_order	distinct_months	verdict
32	2008-06-01	2008-06-01	1	Source CANNOT answer: use AdventureWorksDW

All orders fall on one date, so no month-over-month trend is possible from the source.

2. Fallback: AdventureWorksDW2022 dbo.FactInternetSales

- Revenue = SUM(SalesAmount) per calendar month (excludes tax and freight, like LineTotal).
- MoM change = revenue − LAG(revenue); MoM % = change ÷ previous month.
- 3-month moving average = AVG(revenue) over the current and 2 previous months; shown only when the window has 3 months.
- Profile: orders from 2010-12-29 to 2014-01-28, 60,398 lines, 38 months found vs 38 expected (no gaps).

3. Results (all 38 rows)

Q5 results

month_start	revenue	mom_change	mom_change_pct	moving_avg_3m	is_edge_month
2010-12-01	43,421.04	–	–	–	1
2011-01-01	469,823.91	426,402.88	982.02	–	0

month_start	revenue	mom_change	mom_change_pct	moving_avg_3m	is_edge_month
2011-02-01	466,334.90	-3,489.01	-0.74	326,526.62	0
2011-03-01	485,198.66	18,863.76	4.05	473,785.83	0
2011-04-01	502,073.85	16,875.19	3.48	484,535.80	0
2011-05-01	561,681.48	59,607.63	11.87	516,317.99	0
2011-06-01	737,839.82	176,158.35	31.36	600,531.71	0
2011-07-01	596,746.56	-141,093.26	-19.12	632,089.28	0
2011-08-01	614,557.94	17,811.38	2.98	649,714.77	0
2011-09-01	603,083.50	-11,474.44	-1.87	604,796.00	0
2011-10-01	708,208.00	105,124.51	17.43	641,949.81	0
2011-11-01	660,545.81	-47,662.19	-6.73	657,279.10	0
2011-12-01	669,431.50	8,885.69	1.35	679,395.11	0
2012-01-01	495,364.13	-174,067.38	-26.00	608,447.15	0
2012-02-01	506,994.19	11,630.06	2.35	557,263.27	0
2012-03-01	373,483.01	-133,511.18	-26.33	458,613.77	0
2012-04-01	400,335.61	26,852.61	7.19	426,937.60	0
2012-05-01	358,877.89	-41,457.72	-10.36	377,565.50	0
2012-06-01	555,160.14	196,282.25	54.69	438,124.55	0
2012-07-01	444,558.23	-110,601.91	-19.92	452,865.42	0
2012-08-01	523,917.38	79,359.15	17.85	507,878.58	0
2012-09-01	486,177.45	-37,739.93	-7.20	484,884.35	0
2012-10-01	535,159.48	48,982.03	10.07	515,084.77	0
2012-11-01	537,955.52	2,796.03	0.52	519,764.15	0
2012-12-01	624,502.17	86,546.65	16.09	565,872.39	0
2013-01-01	857,689.91	233,187.74	37.34	673,382.53	0
2013-02-01	771,348.74	-86,341.17	-10.07	751,180.27	0
2013-03-01	1,049,907.39	278,558.65	36.11	892,982.01	0
2013-04-01	1,046,022.77	-3,884.62	-0.37	955,759.63	0
2013-05-01	1,284,592.93	238,570.16	22.81	1,126,841.03	0
2013-06-01	1,643,177.78	358,584.85	27.91	1,324,597.83	0
2013-07-01	1,371,675.81	-271,501.97	-16.52	1,433,148.84	0

month_start	revenue	mom_change	mom_change_pct	moving_avg_3m	is_edge_month
2013-08-01	1,551,065.56	179,389.75	13.08	1,521,973.05	0
2013-09-01	1,447,495.69	-103,569.87	-6.68	1,456,745.69	0
2013-10-01	1,673,293.41	225,797.72	15.60	1,557,284.89	0
2013-11-01	1,780,920.06	107,626.65	6.43	1,633,903.05	0
2013-12-01	1,874,360.29	93,440.23	5.25	1,776,191.25	0
2014-01-01	45,694.72	-1,828,665.57	-97.56	1,233,658.36	1

Revenue dips during 2012 and then grows strongly through 2013. The moving average smooths the monthly swings.

4. Limitations

1. **Different database.** The mart's source (LT) cannot answer this, so results come from DW and must not be reconciled to the mart or to Finance's SUM(SubTotal).
2. **Internet sales only.** DW does not include reseller sales, so it is not the whole business.
3. **Partial first and last months.** Dec 2010 has only 29–31 Dec, and Jan 2014 ends on 28 Jan. The +982.02% (Jan 2011) and −97.56% (Jan 2014) are data artefacts, not real growth or decline. The Feb 2011 and Jan 2014 averages are also distorted. These months are flagged by is_edge_month.
4. **LAG assumes no missing months.** Confirmed here (38 = 38), but it must be re-checked on other data.
5. **Trailing average.** It lags turning points.